

LC 1952: Three Divisors

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Problem Logic & Core Idea

We need to check if a number n has exactly 3 divisors. A number always has 1 and itself as divisors. To have exactly 3, it must have exactly **one** more unique divisor.

Edge Case: If $n < 4$, it can never have 3 divisors (1, 2, 3 only have 1 or 2 divisors). Return false.

Divisor Tracking Strategy

Trigger Flags

- **f1:** Triggers when it gets the *first* divisor apart from 1 and itself.
- **f2:** Triggers on the *second* divisor found (which would be the 4th total divisor).

Checking Divisors

Looping up to $i * i \leq n$:

If $n \% i == 0$, check if it's a perfect square.

Condition 1: $i * i != n$

Matlab paired divisor hoga! If i divides n and isn't the square root, then n/i is also a distinct divisor. Hence automatically it will have ≥ 4 divisors. We can immediately return false;.

Condition 2: $i * i == n$

Matlab 1 hi divisor he. Set `f1 = true` (pehla non-1 and non-itself divisor). If we somehow hit another one, set `f2 = true`.

Final Return Logic:

- `if(f1 && f2) return false; //  $\geq 4$  div`
- `if(f1 && !f2) return true; //  $== 3$  div`
- `if(!f1 && !f2) return false; //  $< 3$  div, prime`

C++ Implementation

Full code handling the mathematical divisor count logic.

```
class Solution {
public:
    bool isThree(int n) {

        if(n<4){
            return false;
        }

        bool f1=false; //trigger when gets first divisor apart from 1 and itself
        bool f2=false; //trigger on second divisor , i.e 4th div

        for(int i=2;i*i<=n;i++){

            if(n%i==0){
                if(i*i!=n){
                    //matlab paired divisor hoga
                    //hence automatically 4 divisor
                    return false;
                }
                else{
                    //matlab 1 hi divisor he
                    if(!f1){
                        f1=true; //pehla non 1 and non itself divisor
                    }
                    else{
                        f2=true; //doosra non 1 and non itself divisor
                    }
                }
            }

        }

        if(f1 && f2) return false; //>=4 div
        else if(f1 && !f2) return true; //==3 div
        else if(!f1 && !f2) return false; //<3 div , prime
        else return false;

    }

};
```